



Lesson 3.8 — The Factor Theorem

Big idea: $(x - c)$ is a factor of $P(x)$ if and only if $P(c) = 0$. That makes finding factors and finding zeros the same task.

Quick review

Statement: $(x - c)$ is a factor of $P(x)$ ■ $P(c) = 0$.

Strategy: if you suspect a zero c , check $P(c)$. If 0, divide $P(x)$ by $(x - c)$ to factor it down.

Rational Root Theorem (helper): any rational root p/q of $P(x)$ (with integer coefficients) has p dividing the constant term and q dividing the leading coefficient.

Factoring cubics and beyond: guess a rational zero from the RRT list, divide it out, factor the resulting quadratic.

Worked example

Worked example. Factor $P(x) = x^3 - 4x^2 + x + 6$.

Possible rational zeros: $\pm 1, \pm 2, \pm 3, \pm 6$. Test $x = -1$: $P(-1) = -1 - 4 - 1 + 6 = 0$ ✓. Synthetic divide $\rightarrow x^2 - 5x + 6 = (x - 2)(x - 3)$.

$P(x) = (x + 1)(x - 2)(x - 3)$.

Warm-ups

1. If $(x - 5)$ is a factor of $P(x)$, what is $P(5)$?
2. List the possible rational zeros of $P(x) = x^3 - 2x - 4$ (RRT).
3. Is $(x + 2)$ a factor of $P(x) = x^3 + 8$?

Practice

1. Factor $x^3 - 7x + 6$ completely.
2. Factor $x^3 + 2x^2 - 5x - 6$ completely (test small candidates).
3. Find all rational zeros of $2x^3 - 3x^2 - 8x + 12$.

Challenge

1. Factor $P(x) = x^4 - 5x^2 + 4$ over the integers. (Hint: substitute $u = x^2$.)
2. $P(x) = x^3 + ax^2 + bx + 2$. Suppose $(x - 1)$ and $(x + 2)$ are both factors. Find a and b , and the third zero.



Answer key — Lesson 3.8

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

- 0.
- $\pm 1, \pm 2, \pm 4$.
- $P(-2) = -8 + 8 = 0 \Rightarrow$ yes.

Practice

- Test $x = 1$: $1 - 7 + 6 = 0 \checkmark$. Synthetic divide $\rightarrow x^2 + x - 6 = (x + 3)(x - 2)$. Answer: $(x - 1)(x + 3)(x - 2)$.
- Test $x = -1$: $-1 + 2 + 5 - 6 = 0 \checkmark$. Synthetic $\rightarrow x^2 + x - 6 = (x + 3)(x - 2)$. Answer: $(x + 1)(x + 3)(x - 2)$.
- RRT candidates: $\pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12, \pm 1/2, \pm 3/2$. Test $x = 2$: $16 - 12 - 16 + 12 = 0 \checkmark$. Divide $\rightarrow 2x^2 + x - 6 = (2x - 3)(x + 2)$. Zeros: 2, $3/2$, -2 .

Challenge

- Let $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4) = (x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2)$.
- $P(1) = 1 + a + b + 2 = 0 \Rightarrow a + b = -3$. $P(-2) = -8 + 4a - 2b + 2 = 0 \Rightarrow 4a - 2b = 6 \Rightarrow 2a - b = 3$. Add: $3a = 0 \Rightarrow a = 0$, $b = -3$. Polynomial: $x^3 - 3x + 2 = (x - 1)(x + 2)(x - 1) = (x - 1)^2(x + 2)$. Third zero is $x = 1$ again (mult 2).